

Lab 6: SQL Joins and Aggregates

CS355/CE373 Database Systems

Fall 2025



Dhanani School of Science and Engineering

Habib University

Copyright © 2025 Habib University

Contents

1	Instructions	2
1.1	Marking scheme	2
1.2	Late submission policy	2
2	Objective	2
3	Query Syntax Examples	2
4	Exercises	3

1 Instructions

- This lab will contribute 1% towards the final grade.
- The deadline to submit this lab is at the end of your lab.
- The lab must be submitted online via CANVAS. The SQL file should be named as *Lab_06_aa01234.sql* where *aa01234* will be replaced with your student id. ***Files which don't follow the appropriate naming convention will not be graded.***

1.1 Marking scheme

This lab will be marked out of 100.

- 50 Marks are for completion of the lab.
- 50 Marks are for Progress and attendance (validated via viva)

1.2 Late submission policy

No late submissions are allowed.

2 Objective

This lab activity is prepared on Northwind Sample Database of SQL Server. The database will be analyzed for the following SQL constructs:

- Joins (Inner / Outer)
- Aggregates

3 Query Syntax Examples

• Join

- select Customers.CompanyName, Customers.CustomerID, Orders.OrderID, Orders.OrderDate
from Customers,Orders
where Customers.CustomerID = Orders.CustomerID
and Country = 'Germany'

• Table Alias

- Select C.customerID, OrderID,OrderDate , E.FirstName
from Orders O
Inner Join Customers C On C.CustomerID = O.CustomerID
Inner Join Employees E On E.EmployeeID = O.EmployeeID
Where C.country = 'Germany'

• Left Outer Join

- Select C.customerID, OrderID,OrderDate, E.FirstName
from Customers C
Left Outer Join Orders O On C.CustomerID = O.CustomerID
Left Outer Join Employees E On E.EmployeeID = O.EmployeeID
Order by O.OrderID

• Cross Join

- Select * from Employees Cross Join Products;
- **Aggregates**
 - Select count(*) from customers
 - Select sum(Quantity) as ‘Total Inventory’ from [Order Details]

4 Exercises

The ERD Diagram for the Northwind Database is as shown in Fig 1.

1. **Retrieve total number of customers do not have a fax number.**
Output: Count of Customers with no fax number.
2. **Retrieve the total number of orders placed in 1997.**
Output: Count of Orders in 1997.
3. **Find the number of customers who are ‘Sales Representative’.**
Output: Count of Customers.
4. **Select total number of products ordered in Order ID 11070.**
Output: Total number of products.
5. **Retrieve total number of customers from the ‘UK’ or ‘USA’.**
Output: Total number of customers.
6. **Retrieve the total number of units of all available items.**
Output: Sum of all units.
7. **Retrieve the worth of all available stock (total number of available units $\times$ their prices).**
Output: Worth of all available products.
8. **Find the total number of Employees who live in London.**
Output: Total number of employees.
9. **Find the total number of Female Employees who are not doctors.**
Output: Total number of employees.
10. **Fetch the following details:**
Output: OrderID, OrderDate, ProductName.
Result contains 2155 rows.
11. **Fetch the following details:**
Output: Order ID, Order Date, Product Name, Customer Name.
Result contains 2155 rows.
12. **Select all orders having products belonging to ‘Seafood’ category.**
Output: OrderID, OrderDate, ProductName, CategoryName.
Result contains 330 rows.
13. **Fetch Customers who have not placed any order.**
Output: CustomerName.
Result contains 2 rows.
14. **Select orders that contain products in either ‘Meat/Poultry’ or ‘Dairy Products’ categories.**

Output: (Order ID)

Result contains 423 row.

15. **Find all possible combinations of employees and customer.**

Output: Employee Full Name, Customer Name.

Note: The Full name is generated by concatenating First Name and Last Name.

Result contains 819 rows.

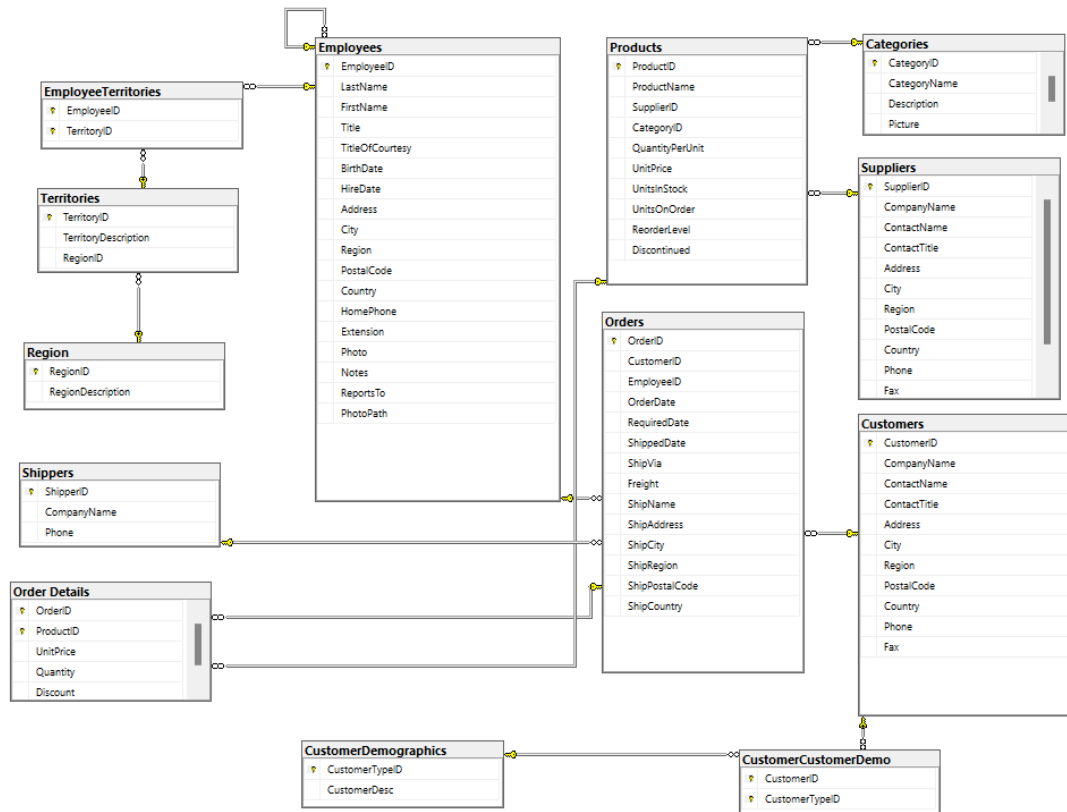


Figure 1: Northwind Database ERD